

10b — Data Replication

Key Concepts

Term	Definition
Replication	Continuously copying data from one DB server to one or more others
Primary (Master)	The server that accepts all writes
Replica (Slave)	Copy of the primary; serves read requests
Replication Lag	Delay between a write on primary and it appearing on replica
Failover	Promoting a replica to primary when the primary goes down
Async Replication	Write confirmed before replica saves — faster, small data loss risk
Sync Replication	Write confirmed only after replica also saves — slower, no data loss

How It Works

Primary-Replica (most common):

- 1. Client sends write → goes to Primary only
- 2. Primary saves the data
- 3. Primary streams changes to Replica(s)
- 4. Reads can be served by any Replica

Replication lag problem:

- 1. User updates profile → write hits Primary ☒
 - 2. User immediately reads profile → hits Replica ❌ (replica not caught up yet)
 - 3. User sees stale data → thinks update failed
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Types of Replication

Type	Description	Used When
Primary-Replica	One writer, many readers	Most web apps
Primary-Primary	Both nodes accept writes	Multi-region writes needed
Sync	Replica confirms before write ack	Banking, payments
Async	Write ack before replica catches up	Social media, most web apps

Replication Lag Solutions

- **Read-your-own-writes** — route user's reads to primary right after their write

- **Sticky sessions** — always route a specific user to the same replica
 - **Wait for replica** — only read from replica after it confirms the latest write
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Industry Examples

- **GitHub** — primary-replica; writes to primary, reads spread globally across replicas
 - **Netflix** — replicates across AWS regions; if us-east-1 fails, eu-west-1 takes over
 - **Twitter** — replication lag caused users to not see their own tweet for ~1 second in early days
 - **Amazon RDS** — managed read replicas with automatic failover to promoted replica
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Interview Questions

Q: What is replication and why do we use it? A: Copying data to multiple servers for high availability (failover) and read scalability (spread read load).

Q: What is replication lag? A: The delay between a write on the primary and it appearing on replicas — can cause users to read stale data right after writing.

Q: Sync vs async replication — when do you use each? A: Sync for critical data (banking) — safer but slower. Async for most web apps — faster but brief data loss risk on primary crash.

Q: What is the difference between replication and sharding? A: Replication = same data on multiple servers (read scale + availability). Sharding = different data on each server (write scale + storage scale).

Q: How do you solve the read-your-own-writes problem? A: Route the user's reads to the primary temporarily after a write, or use sticky sessions to always hit the same replica.